

Google DeepMind Lyria 3: Music Generation Model Outputs 48kHz Stereo with Real-Time Steering and SynthID Watermarking

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The image is a promotional graphic for Google DeepMind Lyria 3. It features a dark blue background with a grid pattern. At the top left, the text "Google DeepMind Lyria 3" is displayed in white and green. Below this, a subtitle reads "AI music generation with real-time steering and SynthID watermarking". The graphic is divided into three main sections by vertical lines: 1. "48kHz" in red text, with sub-points "Studio-quality stereo" and "Full arrangements from scratch". 2. "<2s" in yellow text, with sub-points "Real-time latency" and "Steer music as it plays". 3. "SynthID" in blue text, with sub-points "Inaudible watermark" and "Survives compression". At the bottom left, a line of text states "Text, image, or audio → full multi-instrumental tracks with vocals". In the top right corner, there is a faint, stylized graphic of a musical staff with notes. In the bottom right corner, the "ToKnow.ai" logo is visible.

Google DeepMind
Lyria 3

AI music generation with real-time steering and SynthID watermarking

- 48kHz**
Studio-quality stereo
Full arrangements from scratch
- <2s**
Real-time latency
Steer music as it plays
- SynthID**
Inaudible watermark
Survives compression

Text, image, or audio → full multi-instrumental tracks with vocals

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Figure 1: Google DeepMind Lyria 3

Google DeepMind released Lyria 3, its most advanced music generation model. The model generates 30-second tracks in studio-quality stereo (48kHz) from text, image, or audio inputs. It does not piece together loops. It generates full musical arrangements from scratch, including vocals and multi-instrumental tracks. Lyria 3 is now integrated into the Gemini app. Describe a mood, genre, or set of instruments, and it outputs a finished audio file. Music generation is hard because it is continuous and multi-layered: melody, harmony, rhythm, and timbre all need to stay coherent from the first second to the last.

The real shift is the Lyria RealTime API. Traditional models work like a jukebox: input a prompt, wait for a file. Lyria RealTime generates audio in 2-second chunks as a live stream. Users can steer style, tempo, and instrumentation while the track plays, with under 2 seconds of delay. The Music AI Sandbox builds on this, transforming a hum into a full orchestral arrangement or turning chord progressions into a vocal choir. Every track includes a SynthID watermark embedded in the audio waveform, inaudible to humans but detectable by software even after MP3 compression or analog re-recording. As AI-generated music floods streaming platforms, permanent attribution solves a real trust problem for creators and listeners. Lyria 3 competes directly with Suno and Udio but brings multimodal input and Google's infrastructure advantage.

Sources:

- [Google DeepMind Lyria Technical Details](#)
- [MarkTechPost: Google DeepMind Lyria 3](#)

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